

5(a)(i)	Golgi (body / apparatus / complex) ;	1						
5(a)(ii)	<table border="1"><tr><td>animal cell</td><td>✓</td></tr><tr><td>plant cell</td><td>✓</td></tr><tr><td>bacterial cell</td><td></td></tr></table> ;	animal cell	✓	plant cell	✓	bacterial cell		1
animal cell	✓							
plant cell	✓							
bacterial cell								
5(b)	magnification = scale bar length / actual length ; A $M = I / A$ $M = I / O$ or magnification triangle 14 500 ;; for 29 mm scale bar length A 14 000 (for 28 mm) or 15 000 (for 30 mm) <i>allow one mark only for</i> <i>incorrect answer but correct calculation using scale bar e.g. $29\,000 \div 2$</i> <i>correct answer but units given</i> <i>correct measurement and correct calculation but incorrect conversion</i>	3						
5(c)	<i>mark first two only</i> nucleus ; mitochondria ; chloroplasts ;	2						
5(d)(i)	<i>fluid</i> <i>idea of <u>phospholipid</u> (and protein) molecules, move about / diffuse (within their monolayer) ;</i> <i>mosaic</i> protein (molecules), interspersed / scattered / AW ; A different / AW, proteins ;	2						

Question	Answer	Marks
5(d)(ii)	<i>any four from</i> 1 barrier between cytoplasm and, external environment / AW ; e.g. tissue fluid A protects, organelles from damage / from pathogens, I keeps cell contents in / membrane surrounds the organelles 2 cell signalling or has receptors for, cell signalling substance / hormone / neurotransmitter / AW 3 cell recognition / acts as cell surface antigen ; 4 cell-to-cell adhesion ; 5 site for, enzymes / catalysing reactions ; 6 anchoring the cytoskeleton / AW ; 7 selection of substances that enter or leave a cell ; AW 8 formation of <u>hydrogen bonds</u> with water for stability ; 9 AVP ; e.g. <i>ref. to</i>, changing shape of cell / flexibility of cells e.g. phagocytosis	4